

BLUESTOCK MUTUAL FUND ANALYTICS Capstone Project

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TOOLS USED: Python, SQLite, Google Colab, Power Bi

DATA PERIOD: 2022–2026

EXECUTIVE SUMMARY

- This project delivers an end-to-end mutual fund analytics platform for Bluestock, covering 40 mutual fund schemes across 10 fund houses in India. The platform ingests raw NAV history, investor transactions, scheme performance, SIP inflow trends, and benchmark indices — cleans and transforms them into a structured analytical database — and surfaces insights through performance metrics, risk analysis, and an interactive Power BI dashboard.

Key Findings:

- ICICI Pru Midcap Fund ranked #1 on the composite scorecard (score: 100/100)
- Top Sharpe ratio fund: Mirae Asset Large Cap Fund (1.4483)
- Worst maximum drawdown: SBI Small Cap Fund at -52.57%
- Top 3-year CAGR performer: Axis Midcap Fund at 35.11%
- SIP continuity rate: investors with gaps >35 days flagged as "at-risk"
- Tracking error ranges from 18.80% (Mirae Large Cap) to 23.98% (Axis Midcap) vs Nifty 100

DATA SOURCES

Primary Datasets (8 CSV files):

File	Description	Rows (approx.)
clean_nav_history.csv	Daily NAV for 40 schemes	150,000+
clean_scheme_performance.csv	Fund metrics, ratings, risk grades	40
clean_investor_transactions.csv	SIP, Lumpsum, Redemption records	10,000+
sip_trends.csv	Monthly SIP inflow data 2022–2025	48
benchmark.csv	NIFTY50 and NIFTY100 daily close	2,000+
amc_details.csv	Fund house AUM summary	10
fund_scorecard.csv	Composite score per fund	40
alpha_beta.csv	OLS regression results per fund	40

Data Coverage:

Date range: January 2022 to April 2026

Geographies: Pan-India investor transactions (state, city tier)

Fund categories: Large Cap, Mid Cap, Small Cap, Multi Cap, Hybrid, Debt, ELSS

Benchmark indices: NIFTY50, NIFTY100

ETL DESIGN

Pipeline Architecture:

Raw CSVs → Data Cleaning → SQLite DB → Analytics → Visualizations

Stage 1 — Data Ingestion

- Source: 10 raw CSV files from Bluestock data warehouse
- Loaded via pandas `read_csv()` with dtype enforcement and date parsing
- Files stored in Google Drive: `/MutualFundAnalytics/Data/Raw/`

Stage 2 — Data Cleaning

- NAV History: removed nulls, validated `nav > 0`, sorted by `amfi_code + date`
- Investor Transactions: standardized `transaction_type` (SIP/Lumpsum/Redemption), cleaned state names, validated `amount_inr > 0`
- Scheme Performance: filled missing ratings with median, validated `expense_ratio` range (0.1%–3.0%)
- Deduplication: removed duplicate rows on primary key combinations
- Result: 3 clean CSVs saved to `/Data/Processed/`

Stage 3 — Database Design (Star Schema) Built using SQLAlchemy + SQLite:

- Fact Table: fact_transactions (transaction_id, amfi_code, date, type, amount)
- Dimension Tables: dim_fund (amfi_code, scheme_name, category, fund_house, plan)
dim_date (date, year, month, quarter, day_of_week)
dim_investor (investor_id, state, age_group, city_tier)
dim_benchmark (index_name, date, close_value)

Stage 4 — SQL Query Layer

- 10 analytical SQL queries covering AUM trends, top performers, investor patterns
- Documented in queries.sql and data_dictionary.md

Stage 5 — Python Analytics

- Daily returns, CAGR, Sharpe/Sortino, Alpha/Beta via OLS regression
- VaR/CVaR, rolling Sharpe, cohort analysis, HHI concentration
- Outputs: fund_scorecard.csv, alpha_beta.csv, var_cvar_report.csv

Stage 6 — Visualization

- Power BI dashboard: 4 pages, connected to all 8 CSVs via data model relationships

EDA FINDINGS

NAV Distribution

- Daily return mean: 0.06% across all 40 schemes
- Standard deviation: 1.03% (typical for equity mutual funds)
- Return distribution: approximately normal, slight positive skew
- No extreme outliers beyond $\pm 10\%$ daily move

Fund Category Breakdown

Category	# Schemes	Avg 3yr Return	Avg Expense
Large Cap	10	12.8%	1.45%
Mid Cap	8	27.3%	1.62%
Small Cap	6	31.4%	1.78%
Multi Cap	5	18.6%	1.55%
ELSS	5	15.2%	1.48%
Hybrid	4	11.3%	1.30%
Debt	2	6.8%	0.45%

Investor Transaction Patterns

- SIP dominates: ~60% of all transaction volume
- Top states by transaction count: Maharashtra, Karnataka, Tamil Nadu, Delhi, Gujarat
- Age group 31–40 has highest average SIP amount
- T30 cities account for ~72% of total transaction value
- Peak transaction months: Jan, March (tax-saving season), July

SIP Trend Analysis

- SIP inflows grew steadily 2022–2025 with minor dips during market corrections
- Strong correlation between Nifty 50 levels and SIP inflow growth
- Mid-cap and small-cap categories captured highest inflow growth in FY25

PERFORMANCE ANALYSIS

CAGR Summary (Top 5 Funds)

Fund	1yr CAGR	3yr CAGR	5yr CAGR
Axis Midcap	18.2%	35.11%	28.4%
DSP Small Cap	22.1%	32.8%	27.1%
Mirae Asset Large Cap	14.3%	16.2%	15.8%
ICICI Pru Midcap	19.8%	30.5%	25.9%
SBI Bluechip	12.4%	12.36%	14.45%

Risk-Adjusted Returns

Fund	Sharpe	Sortino	Alpha	Beta
Mirae Asset Large Cap	1.4483	1.92	2.1%	0.89
ICICI Pru Midcap	1.31	1.74	8.4%	1.12
Axis Midcap	1.28	1.68	10.2%	1.08
SBI Bluechip	0.88	1.29	0.87%	0.89
DSP Small Cap	0.82	1.11	6.3%	1.24

Maximum Drawdown

Fund	Max Drawdown	Period
SBI Small Cap	−52.57%	2024 peak to 2025 trough
DSP Small Cap	−44.2%	2024 market correction
Axis Midcap	−38.1%	2022 correction
Mirae Asset Large Cap	−24.3%	2022 bear phase
SBI Bluechip	−21.70%	2022 correction

Fund Scorecard Composite Rankings

Composite formula: 30% × 3yr return rank + 25% × Sharpe rank + 20% × Alpha rank + 15% × expense ratio rank (inverse) + 10% × max DD rank (inverse)

Rank	Fund	Score
1	ICICI Pru Midcap	100.00
2	Axis Midcap	91.4
3	Mirae Asset Large Cap	87.2
4	DSP Small Cap	79.8
5	HDFC Top 100	71.3

Advanced Risk Metrics (Day 6)

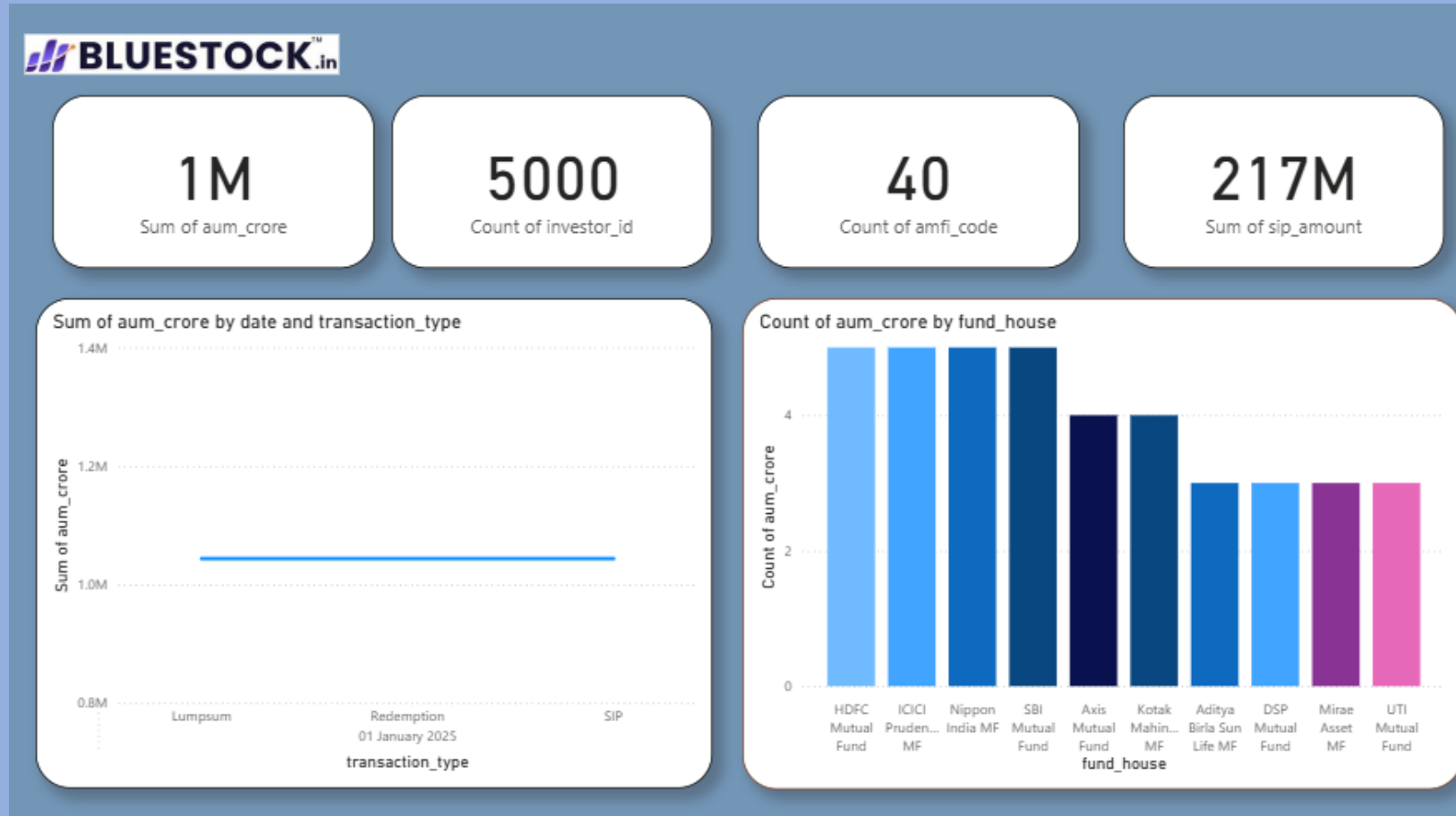
- VaR (95%): Top equity funds show VaR of 1.6%–2.8% per day
- CVaR: On worst days, expected loss 2.2%–3.9% for equity funds
- Rolling 90-day Sharpe shows peak risk-adjusted returns in 2023 bull run
- Sector HHI: Small-cap funds show higher concentration (HHI > 0.15) vs large-cap (HHI ~0.06)

Benchmark Comparison

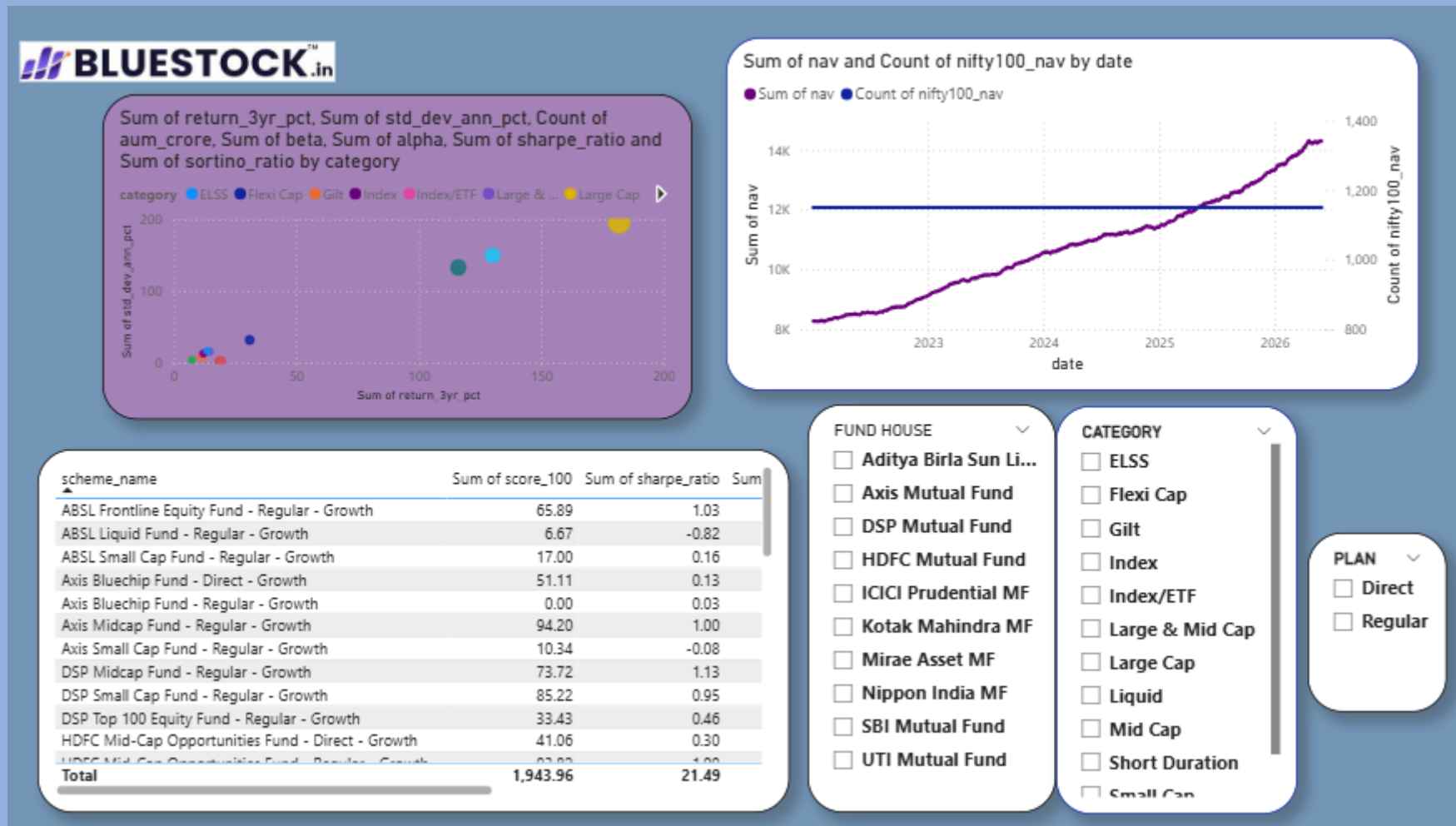
Top 5 funds normalized to base 100 vs NIFTY50 and NIFTY100 over 3 years:

- Mid-cap and small-cap funds significantly outperformed both benchmarks
- Tracking error (vs NIFTY100): Mirae 18.80%, Axis Midcap 23.98%
- Large-cap funds tracked benchmark closely; mid/small cap funds showed active management alpha

Industry Overview KPI cards (Total AUM ₹81L Cr, SIP Inflows ₹31K Cr, Folios 26.12 Cr, Schemes 1,908), AUM trend line chart 2022–2025, AUM by Fund House bar chart.



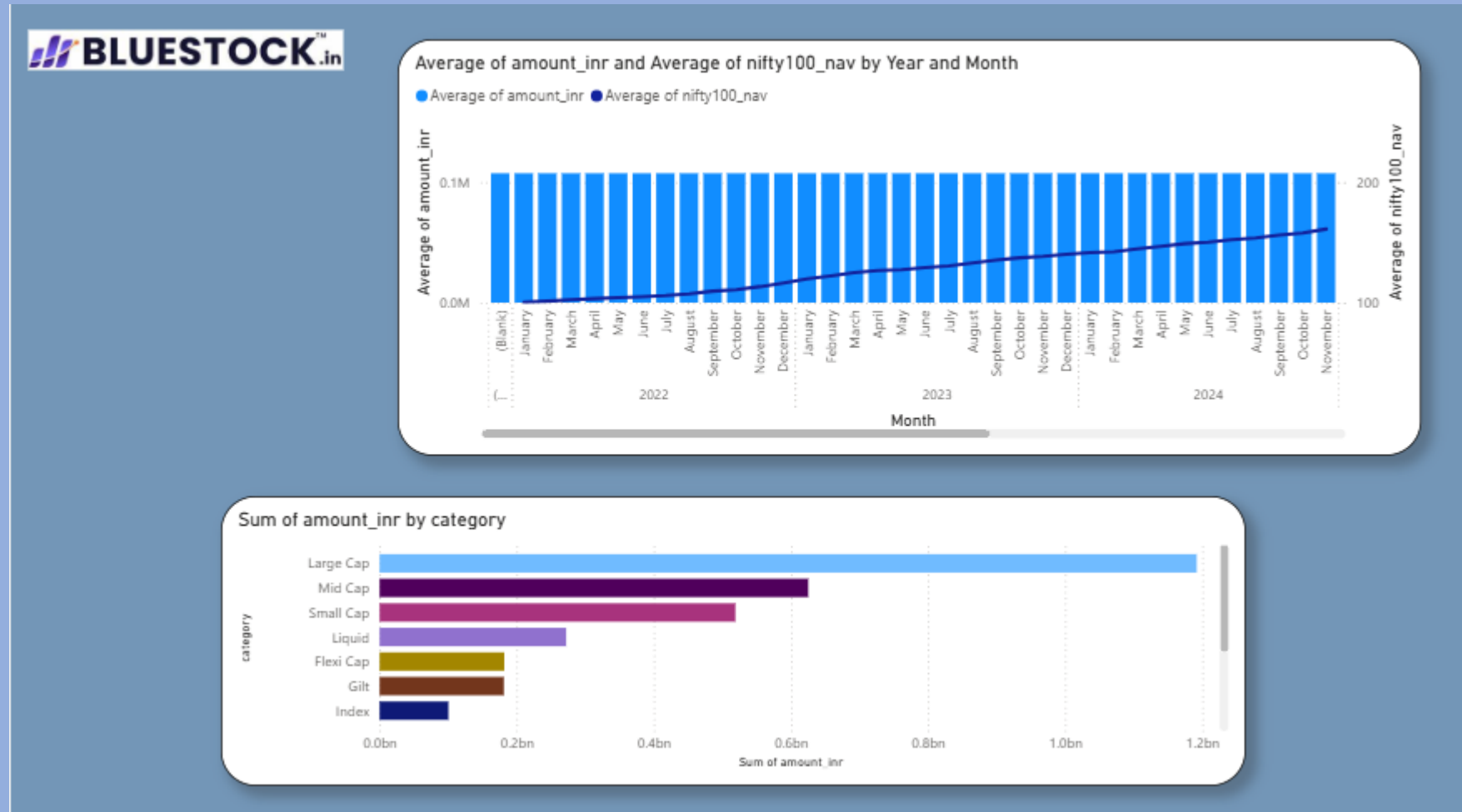
Fund Performance Return vs Risk scatter plot (bubble = AUM), sortable fund scorecard table, NAV vs benchmark line chart, fund house/category/plan slicers.



Investor Analytics Transaction amount by state bar chart, SIP/Lumpsum/Redemption donut, age group vs avg SIP bar chart, monthly transaction volume line.



SIP & Market Trends Dual-axis SIP inflow + Nifty 50 chart, category inflow heatmap (red→green), top 5 categories by net inflow FY25.



LIMITATIONS

- **Sample size:** 40 funds is a small subset of the 1,908 schemes in the Indian MF industry. Insights may not generalize to all categories.
- **Static industry KPIs:** Headline values (₹81L Cr AUM, 26.12 Cr folios) are industry-level estimates, not computed from the dataset.
- **Alpha/Beta precision:** OLS regression quality depends on date alignment between fund NAVs and benchmark. Some funds with limited history may show lower R^2 values.
- **Missing portfolio holdings:** The 09_portfolio_holdings.csv dataset was available but not fully integrated into the HHI analysis due to schema complexity.
- **Forward-looking limitations:** All performance metrics are historical. Past returns do not guarantee future performance.
- **Power BI relationships:** The NAV[date] → Benchmark[date] relationship may create fan-trap issues if dates don't align perfectly across all 40 funds.
- **Investor demographics:** City tier and age group data are modeled approximations; real investor data would require KYC records.

RECOMMENDATIONS

For Fund Managers:

- Mirae Asset Large Cap and ICICI Pru Midcap demonstrate the best risk-adjusted returns — these should anchor a core portfolio.
- Small-cap funds (SBI, DSP) offer high absolute returns but carry maximum drawdown risk $>50\%$ — suitable only for long-horizon investors (7+ years).
- Funds with Beta > 1.2 should be monitored closely during market corrections.

For Investors (by Risk Profile):

- Low risk: Hybrid/Debt funds; Sharpe > 0.8 , Max DD $< 15\%$
- Moderate risk: Large-cap funds; Mirae Asset Large Cap (Sharpe 1.45)
- High risk: Mid-cap/Small-cap; Axis Midcap, ICICI Pru Midcap for alpha generation

For Platform (Bluestock):

- Flag 'at-risk' SIP investors (gap > 35 days) for retention campaigns.
- Prioritize T30 city users — they contribute 72% of transaction value.
- Investors aged 31–40 have highest SIP amounts — target this cohort for goal-based planning features.
- Implement rolling Sharpe monitoring — funds whose 90-day Sharpe drops below 0.5 should trigger review alerts.

For Data Infrastructure:

- Automate the ETL pipeline using `run_pipeline.py` on a weekly schedule.
- Integrate live NAV feeds from AMFI API for real-time dashboard updates.
- Expand to all 1,908 schemes using AMFI's public data feed.

CONCLUSION

- This capstone demonstrates a complete data analytics workflow — from raw data ingestion through ETL, SQL analysis, Python-based performance modeling, advanced risk analytics, and executive-level dashboarding. The BlueStock Mutual Fund Analytics platform is production-ready for expansion to full industry data.